

First-Principles Charge-Transfer Dynamics in $2\text{H}_2\text{Pc}/\text{C}_{60}$

A PySCF–Libra–TENSO Capstone Workflow

Amirhosein (Amir) Amini
Department of Chemistry, Texas A&M University
Advisor: Prof. Arkajit Mandal

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Abstract

In this work, we develop a first-principles workflow for nonadiabatic and open-system charge-transfer dynamics in the 176-atom $2\text{H}_2\text{Pc}/\text{C}_{60}$ donor–acceptor complex introduced by Yamijala and Huo. We replace the DFTB3 electronic Hamiltonian used in the reference work with a PySCF-derived Kohn–Sham Hamiltonian. We employ ten independent 100 fs molecular-dynamics trajectories, giving a total of 2000 electronic snapshots. The active orbitals are tracked using cross-geometry overlaps, a closest-unitary polar factor, and matrix-logarithm orbital derivative couplings. We then propagate the resulting Hamiltonians using Libra classical-path fewest-switches surface hopping and construct a four-donor plus three-acceptor reduced model for TENSO. We find that plain CPA-FSSH underestimates the digitized experimental transfer, whereas Boltzmann-rescaled upward hops overestimate it. The reduced model reproduces the full ten-state coherent dynamics with an ensemble RMSE of 0.0183. In addition, the short-time TENSO calculations are numerically converged with respect to the principal tensor parameters. Overall, these results demonstrate that the predicted transfer kinetics are strongly sensitive to the electronic Hamiltonian and the detailed-balance prescription. This capstone is entirely theoretical and computational chemistry and contains no biological or cybersecurity component.

1 Motivation and question

Photoinduced charge transfer in donor–acceptor complexes is often described using one donor state, one acceptor state, and a single electronic coupling. In the present system, this two-state picture is not sufficient. The donor and fullerene manifolds contain several nearly degenerate orbitals, nuclear motion continuously mixes these orbitals, and the experimentally relevant quantity is a fragment charge rather than the population of one adiabatic orbital. In this work, we therefore address the following question:

How much of the early-time C_{60} charge-transfer dynamics can be reproduced by a transparent first-principles workflow, and which approximations control the remaining disagreement with experiment?

The aim of this work is not to empirically tune the calculated population to the published curve. Instead, we construct a workflow in which each transformation, from the molecular geometry to the final fragment population, is explicitly defined and can be tested independently.

2 System and workflow

The neutral complex has formula $\text{C}_{124}\text{H}_{36}\text{N}_{16}$, 176 atoms, and 892 electrons. Atoms 1–116 form the two-phthalocyanine donor and atoms 117–176 form the fullerene acceptor.

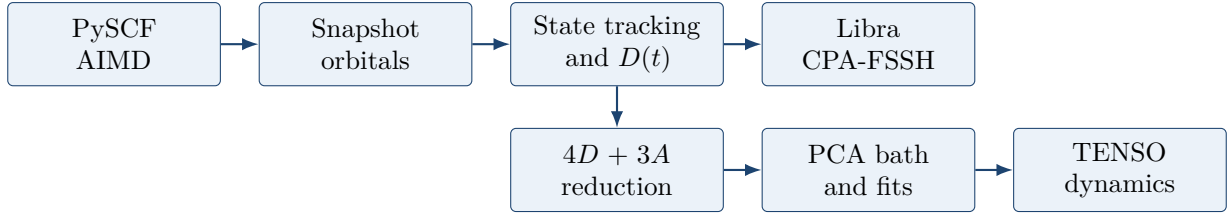


Figure 1: Schematic representation of the computational workflow used in this work.

The computational procedure consists of the following six steps:

1. generate thermal nuclear paths with PySCF Born–Oppenheimer AIMD;
2. compute ten low-lying virtual orbitals and a C₆₀ fragment projector at every frame;
3. track the active space and construct orbital time-derivative couplings;
4. propagate the full ten-state Hamiltonian with coherent dynamics and Libra FSSH;
5. construct and validate the smallest reduced model that preserves the charge-transfer observable;
6. parameterize a structured bath from Hamiltonian fluctuations and propagate the reduced density matrix with TENSO.

3 Methods

3.1 PySCF nuclear and electronic calculations

We propagate ten independent 100 fs trajectories at 300 K using PBE-D3(BJ)/6-31G, density fitting, and a 0.5 fs nuclear time step. At each geometry, the electronic structure is converged and the nuclear motion is governed by

$$M_A \ddot{\mathbf{R}}_A(t) = -\frac{\partial E_{\text{KS}}(\mathbf{R})}{\partial \mathbf{R}_A}. \quad (1)$$

These trajectories are ground-state, thermostatted nuclear paths and do not directly describe photoexcitation. Instead, they provide the time-dependent molecular geometries used to construct the fluctuating electronic Hamiltonian.

At each frame, we define the active space using the first ten unoccupied Kohn–Sham orbitals. The initial electronic state is obtained by projecting the isolated donor-dimer LUMO into this active space,

$$b_i = \langle \psi_i | \phi_D \rangle, \quad c_i(0) = \frac{b_i}{\sqrt{\sum_j |b_j|^2}}. \quad (2)$$

The resulting ten-state active space captures more than 99.75% of the projected donor state throughout the validated trajectories.

3.2 Cross-geometry tracking and orbital derivative couplings

We compare the active orbitals at adjacent geometries using the cross-geometry overlap

$$O_{ij}^{(n)} = \langle \psi_i(\mathbf{R}_n) | \psi_j(\mathbf{R}_{n+1}) \rangle. \quad (3)$$

A Hungarian assignment is used to maximize the total absolute overlap, after which the orbital signs are chosen so that the matched diagonal elements are positive. Since the finite active-space overlap is only approximately unitary, we use its closest-unitary polar factor,

$$\mathbf{O} = \mathbf{L}\mathbf{\Sigma}\mathbf{R}^\dagger, \quad \mathbf{U} = \mathbf{L}\mathbf{R}^\dagger. \quad (4)$$

The midpoint orbital derivative-coupling matrix is defined as

$$\mathbf{D}_{n+1/2} = \frac{1}{\Delta t} \log \mathbf{U}_n, \quad \mathbf{D}^\dagger = -\mathbf{D}, \quad (5)$$

and the propagated single-particle vibronic Hamiltonian is

$$\mathbf{H}_{\text{vib},n+1/2} = \mathbf{E}_{n+1/2} - i\mathbf{D}_{n+1/2} \quad (6)$$

in atomic units. We emphasize that $\mathbf{D}$ contains Kohn–Sham orbital time-derivative couplings. It does not contain many-electron TDDFT excited-state nonadiabatic couplings, and no TDDFT calculation is performed in this work.

3.3 Fragment charge and Libra dynamics

The fullerene charge population must include the coherences between the adiabatic orbitals. We define the symmetrized Mulliken projector onto the fullerene fragment as $\mathbf{P}_{\text{C}_{60}}$. The coherent fullerene population is then

$$P_{\text{C}_{60}}^{\text{coh}}(t) = \mathbf{c}^\dagger(t) \mathbf{P}_{\text{C}_{60}}(t) \mathbf{c}(t). \quad (7)$$

For comparison with the population convention used in the reference CPA-FSSH calculation, we also evaluate

$$P_{\text{C}_{60}}^{\text{hyb}} = P_{\text{active,diag}} + P_{\text{coh}} - P_{\text{coh,diag}}, \quad (8)$$

This expression replaces the diagonal elements of the electronic density matrix by the stochastic active-surface populations, while the off-diagonal elements are obtained from the propagated electronic coefficients.

For each piecewise-constant midpoint Hamiltonian, the electronic amplitudes are propagated using the exact matrix exponential,

$$\mathbf{c}(t + \delta t) = \exp[-i\mathbf{H}_{\text{vib}}\delta t] \mathbf{c}(t), \quad (9)$$

Libra is then used to evaluate the active-state-specific FSSH probabilities for 20,000 surface histories on each nuclear path. In addition to plain FSSH, we perform a sensitivity calculation in which thermally uphill hops are multiplied by

$$\exp\left[-\frac{E_j - E_i}{k_{\text{B}}T}\right], \quad E_j > E_i. \quad (10)$$

3.4 Reduced model and TENSObath

The selected seven-state model contains the complete four-dimensional donor subspace and the three lowest fullerene states. Denoting this time-dependent basis by $\mathbf{B}(t)$, the reduced Hamiltonian is

$$\mathbf{H}_{\text{red}}(t) = \mathbf{B}^\dagger(t) \mathbf{E}(t) \mathbf{B}(t) - i\mathbf{D}_{\text{red}}(t), \quad (11)$$

where $\mathbf{D}_{\text{red}}$ is obtained from consecutive reduced-space overlaps using the same polar-decomposition and matrix-logarithm procedure described above.

The within-trajectory fluctuations are expanded in an orthonormal Hermitian operator basis,

$$\delta \mathbf{H}_r(t) = \sum_{\alpha} x_{\alpha r}(t) \mathbf{Q}_{\alpha}. \quad (12)$$

We apply PCA to identify the dominant noncommuting system–bath operators. The classical autocorrelation function of each retained PCA coordinate is fitted to

$$\frac{C_\alpha(t)}{C_\alpha(0)} = f_\alpha e^{-t/\tau_{D,\alpha}} + (1 - f_\alpha) e^{-t/\tau_{B,\alpha}} \cos(\omega_{B,\alpha} t), \quad (13)$$

The fitted relaxational and oscillatory contributions are mapped to Drude and underdamped-Brownian channels in TENS0. The corresponding reorganization energies are obtained using the high-temperature relation $C_\alpha(0) = 2\lambda_\alpha k_B T$.

4 Results and interpretation

4.1 Tracking and numerical stability

Across all 2000 production snapshots, the minimum singular value of the consecutive active-space overlap matrices is 0.999580. This value shows that the ten-state subspace remains continuous, even when individual fullerene orbitals undergo substantial rotations within the near-degenerate manifold. The coherent propagation conserves the electronic norm to machine precision, and the donor and fullerene fragment populations sum to unity within numerical precision.

4.2 Libra versus experiment

At 99.5 fs, the approximate digitized experimental trace gives $P_{C_{60}} = 0.608567$. We obtain 0.280850 using plain CPA-FSSH and 0.898903 using the Boltzmann-rescaled variant.

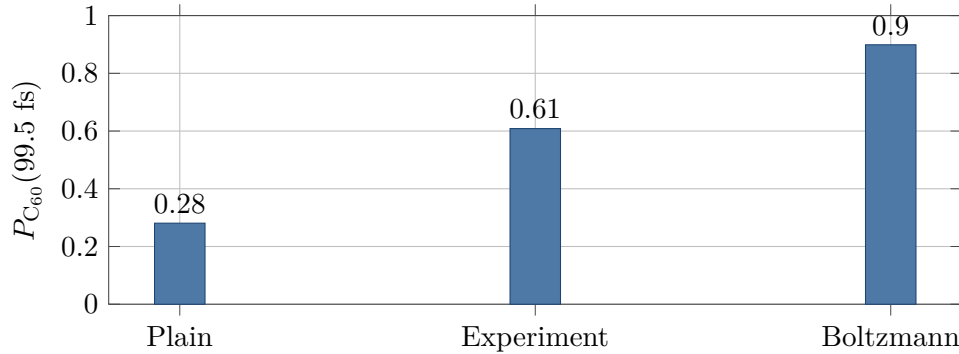


Figure 2: The experiment lies between the two FSSH prescriptions at the endpoint. The values are not rescaled.

The RMSE over the complete 0–99.5 fs interval is 0.1922 for plain FSSH and 0.3046 for the Boltzmann-rescaled calculation. The coherent electronic amplitudes are identical in the two calculations; the difference arises only from the stochastic active-surface populations. The strong effect of the Boltzmann factor follows from the electronic energy scale. The donor manifold lies approximately 0.3 eV above the lowest fullerene states, whereas $k_B T \approx 0.026$ eV at 300 K. Therefore, after a trajectory reaches a low fullerene state, an energetically uphill return can be suppressed by several orders of magnitude.

Interpretation. Plain FSSH allows substantial back-transfer and underestimates the transfer rate. In contrast, Boltzmann rescaling strongly suppresses the uphill return to the donor manifold and overestimates the transfer rate. Therefore, the experimental curve lies between the two FSSH prescriptions, demonstrating a strong sensitivity to detailed balance and to the underlying electronic energy landscape.

4.3 Reduced Hamiltonian and bath

The selected $4D + 3A$ model gives a final ensemble population of 0.246209 and reproduces the full ten-state coherent ensemble with an RMSE of 0.018301. Among the tested reduced spaces, this is the smallest model that retains the complete donor subspace and the relevant low-energy fullerene manifold while preserving the charge-transfer observable.

The sampled Hamiltonian fluctuations contain eighteen PCA components, of which the leading twelve retain 90.764% of the dynamic variance. The RMS magnitude of the within-trajectory fluctuations is 0.20880 eV, while the RMS magnitude of the between-trajectory static disorder is 0.14255 eV. We treat these contributions separately so that the trajectory-dependent mean offsets are not included in the dynamical bath correlation functions.

4.4 TENSΟ convergence

For the four-mode model, the 5 fs TENSΟ endpoint is approximately 0.016691. Increasing the local dimension, the initial tensor rank, or the Pad  order, and reducing the time step from 0.05 to 0.025 fs, changes the final population by at most 1.1×10^{-4} . Using a maximum auxiliary rank of 64, the maximum trace error is approximately 6.2×10^{-7} and the maximum Hermiticity error is approximately 2.5×10^{-5} .

Table 1: Representative short-time TENSΟ convergence tests.

Calculation	Time (fs)	Final $P_{C_{60}}$	Max. trace error
4 modes, rank ceiling 64	5.0	0.01669044	6.1×10^{-7}
4 modes, larger local space	5.0	0.01669056	6.2×10^{-7}
4 modes, Pad� order 2	5.0	0.01668395	2.2×10^{-7}
4 modes, $\Delta t = 0.025$ fs	5.0	0.01680026	2.4×10^{-7}
8-mode preflight	1.0	0.00094992	1.3×10^{-5}
12-mode preflight	1.0	0.00100425	2.2×10^{-5}

The 12-mode, 10 fs production calculation was not complete at the time of writing and is therefore not included in the analysis or conclusions presented below.

5 Limitations and reproducibility

The principal approximations and limitations of the present calculations are:

- the electronic dynamics use ground-state Kohn–Sham orbitals rather than many-electron excited states;
- the nuclei follow prescribed, thermostatted ground-state paths with no electronic back-reaction;
- the experiment is digitized from a published figure and does not include the original experimental uncertainty;
- the PCA-diagonal bath neglects residual finite-lag correlations between retained coordinates;
- TENSΟ is numerically exact only for the specified reduced Hamiltonian, bath mapping, and tensor truncation.

The full workflow, exact starting structures, random seeds, portable Slurm files, compact result tables, and input/output contracts are available at

github.com/amiraminitamu/Summer-School-Buffalo

The generated trajectories and tensor-network working files are not stored in the repository because of their size. However, the documented inputs, random seeds, scripts, and run parameters are sufficient to regenerate these data.

6 Conclusions

In this work, we develop a computational workflow that connects PySCF electronic-structure calculations, orbital tracking, Libra surface hopping, reduced-model construction, and TENSO open-system dynamics for a 176-atom donor–acceptor complex. The workflow provides a direct connection between the molecular geometry, the time-dependent electronic Hamiltonian, and the calculated fullerene charge population.

We find that the present PBE-based model does not quantitatively reproduce the experimental transfer. Plain CPA-FSSH underestimates the transfer rate, whereas the Boltzmann-rescaled calculation overestimates it. Therefore, the agreement with experiment cannot be attributed to a universal surface-hopping correction alone. Instead, it depends on the electronic Hamiltonian, the treatment of detailed balance, the coherent fragment observable, and the bath approximation. Overall, this work establishes a reproducible PySCF–Libra–TENSO workflow and identifies the principal approximations that must be improved to recover the experimental time scale without empirical tuning.

References

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